Evaluating, Transferring, and Ensembling

Modeling the distribution of biodiversity under global climate change

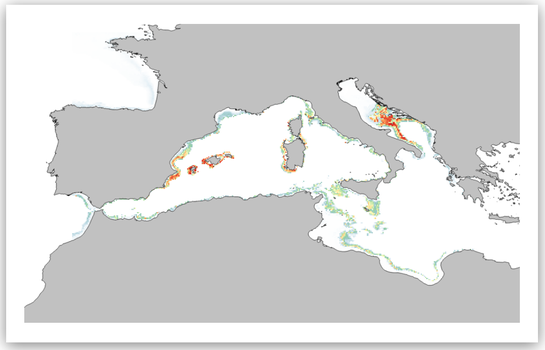
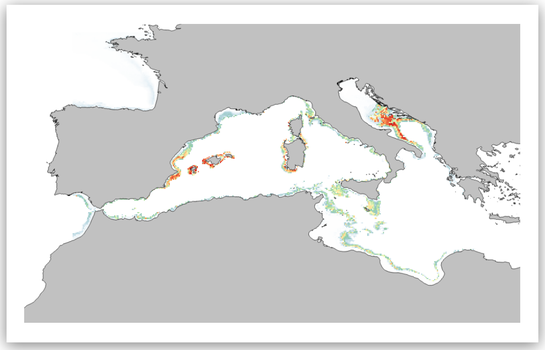
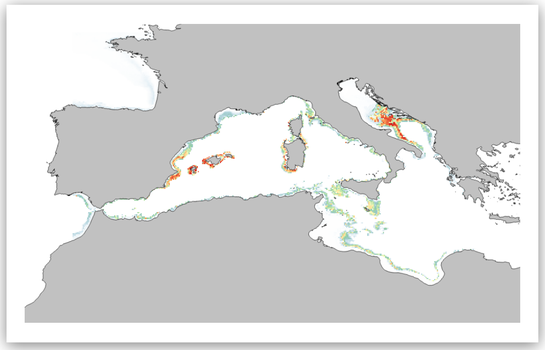
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The Power of Ensemble Modeling

Ensembling models

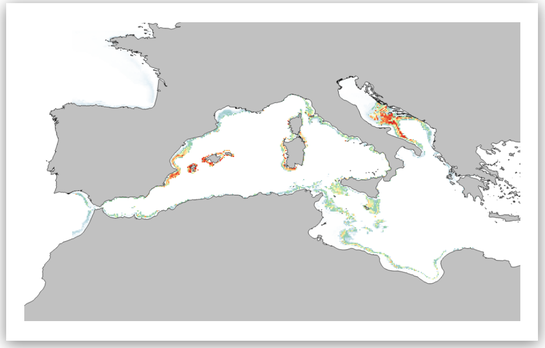
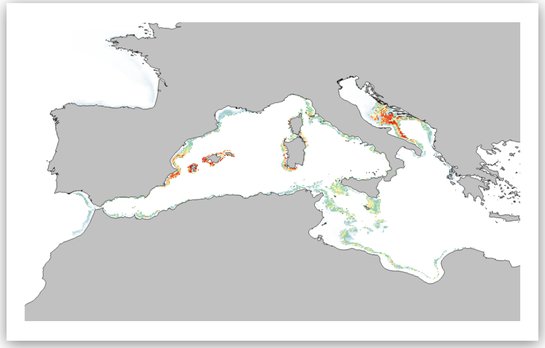
Each algorithm (BRT, GLM) used different approached to fit the data, which brings uncertainty. Instead of using a single algorithm to investigate species distributions, one can conduct an **ensemble experiment**. This **reduces the uncertainty of algorithms by combining their outputs with a general statistics** (e.g., mean).

Techniques: Simple means, or weighted averages (where algorithms that scored higher in cross-validation are given more influence over the final map).

mean( **,**  **) =** 

Mapping uncertainty between models

Ensembles allow us to map where different algorithms agree and where they heavily disagree. By mapping the standard deviation between the models, we can visually identify high-confidence habitats versus highly uncertain regions

SD( , ) = 